
Speaker notes

Plan B - presenting the system live

About 3 minutes, straight through the app

Nuhan runs the app on his laptop and shares the screen. The team talks through each tab while it works live.

DA 2009 - Data Collection Methods II

Assignment 2 (Group Project)

University of Colombo

25ada072 - Pasindu | 25ada073 - Sasini | 25ada141 - Salaama | 24ada076 - Nuhan

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Plan B on one page

Instead of slides, we show the system itself. Nuhan opens and clicks every button from start to finish. Whoever volunteers does the talking - the script marks where a new speaker can take over. If only one person volunteers, they read it all.

Time	Tab	What to press	Who
0:00	Home	Nothing	24ada076 - Nuhan
0:20	Ethics	Read robots.txt	Any member
0:45	Market Data	Get the market data	Any member
1:20	Announcements	Get the announcements, Download, Read the text	Any member
1:55	OCR	Read it	Any member
2:30	Data & Export	Clean and check, Draw the charts, Save	Any member
2:55	Closing	Nothing	Any member, or Nuhan

Remember: keep talking while a page loads. A few seconds of silence feels long; saying 'it's waiting 1.5 seconds between requests - that's our politeness rule' turns the wait into a point in our favour.

Plan A or Plan B? The live demo proves the system works and is the best way to show how it is built, but it needs the laptop, the projector and ideally the internet. The slides are the safe choice. Keep docs/slides.pptx open in the background either way, so you can switch if something fails.

Getting ready

The day before

1. If the project has never been set up on this laptop, double-click 1_SETUP_First_Time_Only.bat and press **Y** when it asks about Tesseract.
2. Start the app and open the **OCR** tab. The line at the top must say that Tesseract is installed and ready - otherwise the OCR stop will not work.
3. Walk through the whole route below twice, with a timer, with the people who will speak.

On the morning of the viva

1. With internet, double-click 3_Collect_Fresh_Data.bat. It takes about 1 to 2 minutes. It saves a fresh copy of every reply and downloads any new documents. If the internet fails during the viva, the app shows these copies instead.

Ten minutes before

1. Double-click 2_START_The_App.bat. The browser opens the app by itself at `http://127.0.0.1:7860`. Keep the black window open - make it small, but do not close it.

2. Press **Light theme** at the top right - it is easier to read on a projector.
3. Zoom the browser to about 110% (**Ctrl** and **+**) so the back of the room can read it.
4. Close other browser tabs and turn on **Do not disturb**, so no messages pop up while you share.
5. Have these open in the background for the questions: VS Code with the project, File Explorer at `data\exports`, and `docs/slides.pptx`.
6. Leave the app on the **Home** tab. Do not press any buttons yet - the demo shows them working for the first time.

Be careful: the numbers on screen - the number of companies, the index, how many rose and fell - change every trading day. On a weekend or a holiday the market is closed and the app shows the last trading day. Say what you see, not what these notes say.

The live script

Seven stops, about 3 minutes. For each stop: what Nuhan presses, how long it takes, what the speaker says, and what to point at.

Stop 1 - Home (0:00)

Who speaks	24ada076 - Nuhan
Nuhan presses	Nothing - the app is already open on the Home tab.
Wait	-

Say

Good morning. I'm Nuhan, and this is our DA 2009 project: collecting data from the Colombo Stock Exchange. Instead of slides, we'll show you the system running live.

Here's who built each part - Pasindu, Sasini, Salaama and me - and every tab is labelled with its member.

Point to

- The names in the title bar.
- The **Who did what** table.

A new speaker can take over here.

Stop 2 - Ethics (0:20)

Who speaks	Any member
Nuhan presses	The Ethics tab, then Read robots.txt .
Wait	1 to 2 seconds

Say

This is Nuhan's part, and it comes first. Before any request, the app reads CSE's robots.txt - it blocks just one folder, cgi-bin, which we never use.

We wait 1.5 seconds between requests, though CSE doesn't ask us to, and our User-Agent says who we are. Every request we've ever sent is logged here.

Point to

- The Disallow: /cgi-bin/ line in the robots.txt box.
- **Our settings**: the delay and the User-Agent.
- Scroll down to **Every request we have sent** - the number of requests recorded.

Be careful: If the robots.txt box is empty and Our settings says 'could not read robots.txt (no internet?)', there is no internet. Say: "There's no internet here, so the app sends nothing at all and shows the copies we saved this morning." Then carry on - every other stop still works.

Stop 3 - Market Data (0:45)

Who speaks	Any member
Nuhan presses	The Market Data tab, then Get the market data .
Wait	about 8 seconds

While it loads, say: "It's waiting 1.5 seconds between each request - that's the politeness rule working."

Say

This is Pasindu's part. The CSE page is built by JavaScript, so a plain download found only 24 characters of text.

In the browser's Network tab we found the addresses the page itself uses, like tradeSummary. They send JSON, and pandas makes this table - every company from one request, plus the index, and how many rose and fell.

Point to

- The row of figures: ASPI index, companies, rose, fell.
- The **Share prices** table.
- Open **The addresses we read the data from** - each address, and whether it is GET or POST.
- The **Messages** box - one line per request, status 200.

Colombo Stock Exchange — Web Data Collection

DA 2009 - Assignment 2 - 25ada072 - Pasindu | 25ada073 - Sasini | 25ada141 - Salaama | 24ada076 - Nuhan

Home Ethics **Market Data** Companies Announcements OCR Data & Export

written by 25ada072 - Pasindu

Today's trading for every company listed on the exchange, plus the market index and the sector indices.

Get the market data

MARKET
Market Closed

ASPI INDEX
21,037.35
-25.54 today

COMPANIES
284
traded today

ROSE
96

FELL
113

TURNOVER
Rs 0.49 bn
total traded

Share prices Sectors

id	name	symbol	logoUrl	quantity	percentageChange
284	ABANS ELECTRICALS PLC	ABAN.N9999	upload_logo/642_1603172106.jpeg	40	1.535140321429904
1046	ABANCO FINANCE PLC	ABANCO.N9999	upload_logo/1046_1601600773.jpeg	9	1.107404700410161

What the Market Data tab looks like after the button is pressed (light theme).

A new speaker can take over here.

Stop 4 - Announcements (1:20)

Who speaks	Any member
Nuhan presses	The Announcements tab, then Get the announcements . Set How many to download to 3 and press Download . Then scroll down and press Read the text (the first document is already chosen).
Wait	about 2 seconds for each button

Say

This is Sasini's crawler. It reads the list of circulars, finds each PDF on CSE's file server, and downloads a few at a time - ones we already have just say Already saved.

Then Nuhan's reader opens a document: pdfplumber reads the text page by page, and PyPDF2 gives the file details. But one document is a scan, with no text at all.

Point to

- The **Downloaded** table - 'Already saved'.
- **The documents we have** - the Type column says 'Scanned image' for the de-listing notice.
- **Page by page** and **File details** after Read the text.

Stop 5 - OCR (1:55)

Who speaks	Any member
Nuhan presses	The OCR tab. The de-listing notice (it starts with 14364) is chosen already. Leave Image detail at 200 and press Read it .
Wait	about 5 to 10 seconds

While it loads, say: "It's turning the page into an image at 200 dots per inch."

Say

This is Salaama's OCR, in the four steps from the lecture. The page becomes an image, and OpenCV turns it grey, then pure black and white, so the letters have sharp edges.

Tesseract reads the letters - here's the text - and gives each word a confidence score, so we know which words to check by hand.

Point to

- The three pictures, left to right.
- **Text read from the image**.
- **Least confident words** - the ones to check.

Be careful: If the tab shows installation steps instead of pictures, Tesseract is not installed on this laptop. Say so, and show slide 7 of docs/slides.pptx instead.

A new speaker can take over here.

Stop 6 - Data & Export (2:30)

Who speaks	Any member
Nuhan presses	The Data & Export tab, then Clean and check , then Draw the charts , then Save .
Wait	about 1 second for each button

Say

Website data is never ready to use. Sasini's cleaning turns dates from huge numbers into Sri Lanka time, and the checks confirm there are no negative prices and no company twice.

Salaama's charts show the biggest movers, turnover and sectors. Save writes CSV, Excel and JSON, with CSE credited inside every file.

Point to

- **Before and after** - Date columns go from 0 to 1.
- **Checks** - every row says OK.
- The three charts.
- **Files saved so far**.

Be careful: If it says 'Get the market data first', the Market Data button was not pressed - go back one stop and press it.

Stop 7 - Closing (2:55)

Who speaks	Any member, or Nuhan
Nuhan presses	Nothing. (You can click back to Home .)
Wait	-

Say

That's the whole system, running live. Thank you - we're happy to take questions, and we can show any part in more detail.

Extras for the questions

The 5 minutes of questions are on what each member did. These are ready-made things to show when a question comes up. Nothing here is part of the 3 minutes.

4_Run_Each_Members_Part.bat opens a menu that runs any member's file on its own and prints what it does:

Member	Menu options
24ada076 - Nuhan	1 Ethics 2 Static scraper 3 PDF reader
25ada072 - Pasindu	4 API client 5 Selenium
25ada073 - Sasini	6 Crawler 7 Cleaner 8 Scrapy spider
25ada141 - Salaama	9 OCR 10 Storage 11 Analysis

Options 3 and 9 need some PDFs, so run option 6 first on a new computer.

24ada076 - Nuhan - Ethical scraping, BeautifulSoup and PDFs

The request log. Ethics tab, **Show the request log**.

Say: Every request we have sent: the time, the address, the status and the delay we waited.

The ethics code. VS Code: src/ethics.py, the fetch() function.

Say: One function for every request: check robots.txt, wait, send, record - and use a saved copy if the internet fails.

The plain download. 4_Run_Each_Members_Part.bat, option 2.

Say: requests and BeautifulSoup get 17,898 bytes of HTML but only 24 characters of text and no tables - the page is built by JavaScript.

The PDF reader. 4_Run_Each_Members_Part.bat, option 3.

Say: Every saved PDF, page by page: text-based, scanned or hybrid.

25ada072 - Pasindu - Getting the data, and Selenium

One company. Companies tab, **Load the company list**, choose SAMP.N00000 - SAMPATH BANK PLC, **Get details**.

Say: This address needs the company symbol sent with a POST request. It shows the price, market value, and the year's high and low.

Several companies. Same tab: set **How many companies** to 5, **Collect them**.

Say: It takes several seconds because every company is a separate request, and each one waits 1.5 seconds first.

The addresses. Market Data tab, **The addresses we read the data from**.

Say: Every CSE address we use, and whether it is GET or POST.

Selenium. 4_Run_Each_Members_Part.bat, option 5.

Say: Opens a hidden Edge browser, waits for the JavaScript, and finds the table - but it takes about 5 seconds for one page.

25ada073 - Sasini - Crawling for documents, and cleaning

The Scrapy spider. Double-click `6_Run_The_Scrapy_Spider.bat`.

Say: The same crawl done with Scrapy. At the end, `robotstxt/request_count` shows it read `robots.txt` first, and `item_scraped_count` how many documents it found.

The crawler code. VS Code: `src/crawler.py`, `download_documents()`.

Say: It skips files we already have, and checks the first four bytes say `%PDF` before saving.

The cleaning. **Data & Export** tab, **Before and after**, then `src/cleaner.py`, step 1.

Say: Dates arrive as milliseconds in UTC; we turn them into Sri Lanka time, so trades show between 09:30 and 14:30.

25ada141 - Salaama - OCR, saving the files, and the charts

OCR detail. **OCR** tab: move **Image detail** to 100, then 300, pressing **Read it** each time.

Say: Fewer dots per inch is faster but blurrier; more is sharper but slower. 200 is the balance we chose.

The source inside a file. File Explorer: `data\exports`, open the newest `cse_share_prices_...csv` in Notepad.

Say: The first lines start with `#` and credit the Colombo Stock Exchange. In the Excel file, it is the second sheet, **Source**.

The charts. **Data & Export** tab: hover over any bar.

Say: Each bar shows its exact value. Green up, red down - and every bar also says up or down, so colour is never the only clue.

If something goes wrong

What happens	What to do
No internet. The robots.txt box is empty, and the Messages box says 'Using the saved copy of ... instead'.	Keep going - every tab works from the copies saved that morning. Say: "There's no internet here, so the app sends nothing and shows the copies we saved earlier." New documents cannot be downloaded; the ones we have say 'Already saved'.
Slow internet. A button seems stuck.	Keep talking. Each request gives up after 30 seconds, and the Messages box shows what is happening.
OCR shows installation steps.	Tesseract is not installed. Say so, and show slide 7 of docs/slides.pptx instead.
A table stays empty, or a red error appears.	Press the button once more. If it still fails, move on: "This part talks to the live website, which does not always answer."
'Get the market data first' on the Data & Export tab.	The Market Data button was skipped. Go back, press it, return.
The page will not load, or the black window was closed.	Double-click 2_START_The_App.bat again. If the app is still running, it just opens the browser.
Everything fails.	Switch to Plan A: open docs/slides.pptx and press F5.